A PROJECT TITLE

ON

ANTI-MONEY LAUNDERING DETECTION USING GRAPH NEURAL NETWORKS FOR FINANCIAL TRANSACTION ANALYSIS

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

Name: G V MADHAV 2520030568

Name: G VAISHNAVI REDDY 2520030247

Name: K RICHIE KIRAN 2520030498

Under the Esteemed guidance of

Mrs. JYOTHSNA DATTI

ASSISTANT PROFESSOR, CSE

CSE

Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

Name of Title: Anti-Money Laundering Detection Using Graph Neural Networks (GNN)

1. Abstract:

Money laundering has become an increasingly sophisticated financial crime that enables illegal activities such as fraud, terrorism financing, corruption, and cybercrime. Conventional Anti-Money Laundering (AML) systems primarily rely on predefined rules and manual investigation, making them less effective in identifying complex and evolving laundering patterns. With the rapid growth of digital banking, cryptocurrency transactions, and interconnected financial networks, there is a growing need for intelligent detection methods capable of analyzing relationships between entities rather than evaluating transactions in isolation. Recent advancements in Graph Neural Networks (GNNs) have demonstrated significant potential in addressing this challenge by representing financial ecosystems as graphs, where accounts are modeled as nodes and transactions as edges.This study presents a comprehensive overview of existing research on AML detection using GNN-based techniques, including Graph Convolutional Networks (GCN), Graph Attention Networks (GAT), GraphSAGE, and other graph learning approaches. The literature indicates that these models effectively capture structural and contextual information within transaction networks, allowing them to identify suspicious activities that traditional machine learning methods often overlook. Widely adopted benchmark datasets, such as the Elliptic Bitcoin Dataset and AMLSim, have further accelerated research by providing realistic environments for evaluating detection performance. Despite promising improvements in accuracy, precision, and recall, several challenges remain, including severe class imbalance, limited availability of real-world labeled data, scalability to massive transaction graphs, privacy constraints, and the need for explainable predictions that support regulatory compliance.The review also highlights emerging research directions such as heterogeneous graph learning, temporal graph neural networks, self-supervised representation learning, and explainable artificial intelligence, which aim to improve both detection capability and model transparency. Overall, Graph Neural Networks represent a transformative approach to AML by leveraging the relational nature of financial transactions, enabling earlier identification of illicit activities while reducing false positives. Continued research integrating advanced graph learning techniques with scalable and interpretable frameworks is expected to play a crucial role in strengthening financial security and supporting more effective anti-money laundering systems in modern digital economies.

2. INTRODUCTION:

Money laundering is one of the major financial crimes affecting banks, financial institutions, businesses, and digital payment platforms around the world. It involves disguising money obtained from illegal activities to make it appear to have originated from legitimate sources. As financial systems have become increasingly digital, criminals have developed more complicated methods of moving money through multiple accounts, transactions, organizations, and geographical locations. These activities can involve many interconnected transactions, making suspicious behavior difficult to identify using conventional techniques.

Anti-Money Laundering (AML) systems are designed to monitor financial activities and identify transactions or customers that may be associated with illegal activity. Traditional AML systems generally depend on predefined rules, transaction thresholds, customer profiles, and manually designed risk indicators. Although rule-based systems are useful for identifying known patterns, they have difficulty detecting new and complex laundering techniques. They can also produce a significant number of false positives, increasing the workload for investigators and financial institutions.

Machine learning techniques have been introduced to improve AML detection by learning patterns from historical transaction data. Classification and anomaly-detection algorithms can identify transactions that differ from normal financial behavior. However, many conventional machine learning models treat transactions or accounts as individual data points and do not fully consider the relationships between them. In real-world financial systems, these relationships are important because money laundering may involve groups of accounts transferring money through a series of transactions rather than a single suspicious transaction.

Graph-based machine learning provides a suitable representation for such interconnected financial activities. A financial transaction network can be represented as a graph in which accounts, customers, businesses, or cryptocurrency wallets are nodes, while transactions between them are edges. Additional information such as transaction amount, time, transaction type, and account characteristics can be associated with nodes or edges. This allows the system to examine not only individual behavior but also connections and surrounding transaction patterns.

Graph Neural Networks (GNNs) are deep learning models designed to work with graph-structured data. A GNN learns information from a node's features as well as information propagated from neighboring nodes. This makes GNNs useful for AML detection because suspicious financial behavior can often be identified through relationships between multiple entities. A group of accounts that repeatedly transfer funds through a particular sequence may exhibit a suspicious structure even when individual transactions appear normal.

Different GNN architectures, including Graph Convolutional Networks (GCN), Graph Attention Networks (GAT), GraphSAGE, and other graph-learning approaches, have been explored for financial fraud and AML detection. These approaches can learn complex patterns from transaction networks and potentially identify suspicious entities that traditional rule-based systems may overlook. Research using datasets such as the Elliptic Bitcoin Dataset and AMLSim has demonstrated the usefulness of graph-based approaches for studying illicit transaction patterns.

However, applying GNNs to AML detection also presents several challenges. Financial transaction datasets are generally highly imbalanced because legitimate transactions greatly outnumber confirmed illicit transactions. Real-world labeled AML data is difficult to obtain because of privacy, security, and regulatory restrictions. In addition, transaction networks can become extremely large and continuously change over time, creating challenges related to computational efficiency and scalability. Explainability is also important because financial institutions need to understand why a transaction or account has been classified as suspicious.

The proposed project focuses on developing an intelligent AML detection approach using Graph Neural Networks to analyze relationships within financial transaction networks. The system aims to process transaction data, construct a graph representation, learn patterns from connected entities, and identify potentially suspicious transactions or accounts. Performance can be evaluated using metrics such as precision, recall, F1-score, and accuracy, with particular attention given to reducing false positives and handling class imbalance.

The overall objective of this project is to demonstrate how GNN-based learning can complement conventional AML systems by providing a relationship-aware approach to suspicious activity detection. Instead of relying only on individual transaction characteristics, the proposed approach considers the broader structure of the financial network. This can help identify coordinated and hidden transaction patterns and provide a stronger foundation for intelligent financial crime detection. Future improvements may include temporal GNNs, heterogeneous graphs, explainable AI, and real-time transaction monitoring.

3. SOFTWARE REQUIREMENTS:

Python 3.x, Jupyter Notebook / VS Code	Used for developing, training, and testing the AML detection system.
PyTorch & PyTorch Geometric (PyG)	Used to build and train the Graph Neural Network model.
Pandas, NumPy & Scikit-learn	Used for data preprocessing, analysis, and model evaluation.
NetworkX & Matplotlib	Used for transaction graph construction, analysis, and visualization.

Signature of the Guide

Course Instructor